

12_For_Loop

September 15, 2026

```
[2]: fruits=["apple","banana","cherry"] #Lists conatining frutis
      print(fruits)
      print(type(fruits))
```

```
['apple', 'banana', 'cherry']
<class 'list'>
```

```
[3]: for i in fruits:
      print(fruits)
```

```
['apple', 'banana', 'cherry']
['apple', 'banana', 'cherry']
['apple', 'banana', 'cherry']
```

```
[4]: for i in fruits:
      print(i)
```

```
apple
banana
cherry
```

```
[ ]: # After seeing the above two examples. we can clearly see that below
      ↪understandings
      # for --> keyword indicating the start of the loop
      # Iterable --> The sequence or Iterable object being looped for. Example here
      ↪fruits list is iterable in above example
      # variable/Iterator --> It takes on the value of each item in the iterable
      ↪during each iteration.
```

```
[6]: # Range Function
      # We can generate a sequence of numbers using range() function.
      # Range(10) generates numbers from 0 to 9 (10 numbers).
      # We can also define the start, stop and step size as well as per our needs.
      # default step size is 1 if we don't specify.
      # Range fucntions generally doesn't store all the values in memory as it would
      ↪be inefficient.
      # so it remembers the start, stop and step size and generates the next number
      ↪on the go.
```

```
[7]: for i in range(5):  
      print(i)
```

```
0  
1  
2  
3  
4
```

```
[9]: for i in range(0,20,5):  
      print(i)
```

```
0  
5  
10  
15
```

```
[ ]: # Using length and range together instead of index values to fetch iterable,  
      ↪ values using for loop  
print(fruits)  
for i in range(len(fruits)):      # here len gives 3 and range(3) gives 0,1,2,  
      ↪ and we can use those values as index in each iteration  
    print(fruits[i])
```

```
['apple', 'banana', 'cherry']  
apple  
banana  
cherry
```

```
[15]: # Enumerate merges direct iterator in for loop and above length and range combo,  
      ↪ together  
# Enumerator generates two items one is index and other is value as Index, Value  
print(fruits)  
print(enumerate(fruits))      # returns Iterable holding result as Index,  
      ↪ Value  
for index, value in enumerate(fruits):  
    print("At position {}, we have fruit named {}".format(index,value))
```

```
['apple', 'banana', 'cherry']  
<enumerate object at 0x7b16d3292520>  
At position 0, we have fruit named apple  
At position 1, we have fruit named banana  
At position 2, we have fruit named cherry
```

```
[16]: # For loop with Strings  
name="Hanumanth"  
  
for i in name:  
    print(i)
```

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```
[ ]: # if else condition inside a for loop
nums = [1,2,3,4,5,6]

for n in nums:
    if n%2 == 0:
        print("Number {} is even number".format(n))
    else:
        print("Number {} is odd number".format(n))
```

Number 1 is odd number
Number 2 is even number
Number 3 is odd number
Number 4 is even number
Number 5 is odd number
Number 6 is even number

```
[22]: students=[{"name":"Hanu","Percentage":90},
                 {"name":"Allada","Percentage":80},
                 {"name":"Rao","Percentage":70},
                 {"name":"Hanumanth", "Percentage":30}]
print(students)
print(type(students))
print(type(students[0]))
```

```
[{'name': 'Hanu', 'Percentage': 90}, {'name': 'Allada', 'Percentage': 80},
{'name': 'Rao', 'Percentage': 70}, {'name': 'Hanumanth', 'Percentage': 30}]
<class 'list'>
<class 'dict'>
```

```
[27]: # Combination of dictionaries, list, for loop and if else if
for stud in students:
    stud_name=stud["name"]
    stud_Perc=stud["Percentage"]

    if stud_Perc >= 90:
        print(f"{stud_name}% is {stud_Perc} and he is excellent")
    elif stud_Perc >=80 and stud_Perc < 90:
        print(f"{stud_name} % is {stud_Perc} and he is Good")
```

```

elif stud_Perc >= 70 and stud_Perc < 80:
    print(f"{stud_name} % is {stud_Perc} and he is Average")
else:
    print(f"{stud_name} % is {stud_Perc} and he is Poor")

```

```

Hanu% is 90 and he is excellent
Allada % is 80 and he is Good
Rao % is 70 and he is Average
Hanumanth % is 30 and he is Poor

```

```

[42]: # Prime numbers falls between given two range of numbers
start_num = -3
end_num = 50

for num in range(start_num, end_num+1): # to include the end_num, added +1 to
    ↪end_num as range considers before the end_num
    if num > 1:
        prime_cnt=0
        for i in range(2,num):
            if num%i==0:
                prime_cnt += 1
                break # we can keep this to stop if its already divided by
    ↪other nums to save some iterations/processing time.
        if prime_cnt == 0:
            print(f"{num} is Prime Number")
        else:
            print(f"{num} is negative number or less than Prime minimum value")

```

```

-3 is negative number or less than Prime minimum value
-2 is negative number or less than Prime minimum value
-1 is negative number or less than Prime minimum value
0 is negative number or less than Prime minimum value
1 is negative number or less than Prime minimum value
2 is Prime Number
3 is Prime Number
5 is Prime Number
7 is Prime Number
11 is Prime Number
13 is Prime Number
17 is Prime Number
19 is Prime Number
23 is Prime Number
29 is Prime Number
31 is Prime Number
37 is Prime Number
41 is Prime Number
43 is Prime Number
47 is Prime Number

```